

Rock Paper Scissors Simulator!

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Abstract—The Rock Paper Scissors Simulator is a web-based application that allows users to play the famous game “Rock Paper Scissors” against a computer opponent. The application provides an interactive interface where users can select rock, paper, or scissors and receive immediate feedback on the result of each round. The system displays what the computer chose via text and an image, what the player chose via text, and the results of the round. This paper describes the development of the application, including the design, functionality, and future improvements.



1. Introduction

The purpose of developing the Rock Paper Scissors Simulator is to demonstrate the fundamentals of web application development through an interactive and familiar game. Simple games like Rock Paper Scissors provide an engaging way to develop user input handling, conditional logic, and dynamic updates in the web environment.

The application allows users to select between rock, paper, and scissors and play against a computer-generated opponent. After the user makes their choice, the computer opponent makes their choice randomly, and the application determines the outcome based on standard game rules and then displays the results of the round.

The interface is designed to be simple, responsive, and intuitive, allowing users to play each round quickly. The goal of this project is to reinforce core programming

concepts while creating a functional and interactive experience.

2. Background

There are many different online implementations of Rock Paper Scissors, all ranging from simple text-based versions to something more advanced and complex. Websites and mobile applications often include variations with score tracking, animations, and even different mode types.

This project focuses on a simplified version of the game to emphasize the core programming concepts rather than some more advanced features.

3. Proposed Application

The application was developed using HTML, CSS, and JavaScript. The interface was created using standard HTML and styled with CSS. JavaScript was used to handle the logic and dynamics, such as user input, generating the computer's choice, and determining the game's result.

When the user opens the application, they are presented with three options: Rock, Paper, and Scissors. Each option is represented by a button. When the user selects an option, the application generates a random choice for the computer opponent.

The result of the round is based on the game's rules:

- Rock beats Scissors
- Scissors beats Paper
- Paper beats Rock

The result is displayed immediately on the screen.

The game logic is implemented using conditional statements. After both the user and computer opponent make their selections, the program then compares the two values to determine the result.

If both the user and the computer opponent select the same choice, the result is a tie. Otherwise, the application checks what the user chose. Then checks whether the computer opponent has the winning choice; if not, then the player wins the round.

The interface includes buttons for each selection and a display area that shows the user's choice, the computer's choice, and the result of the round.

4. Experimental Results

Here is the initial start-up screen. The game starts immediately and prompts the user to make their selection of Rock, Paper, or Scissors.

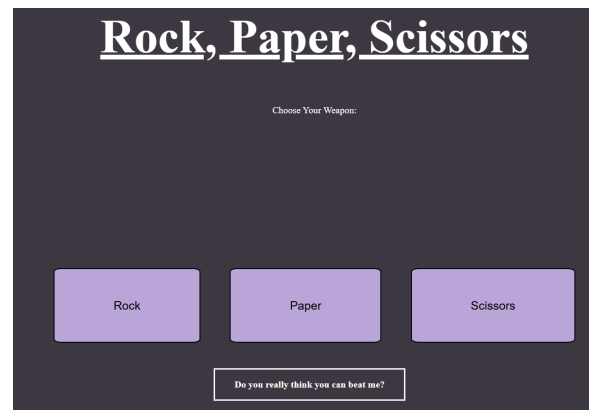


Figure 1: Initial Screen

When the user selects "Rock", the system generates a random choice for the computer opponent. If the computer opponent selects "Scissors", the system correctly displays a win for the user.

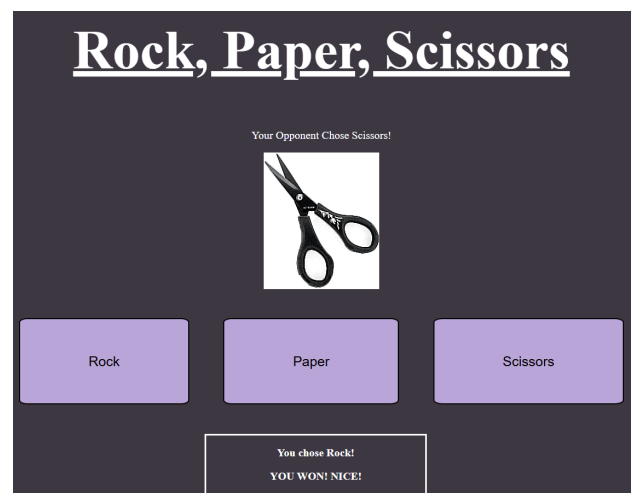


Figure 2: User Selects Rock

Underneath the selections, the player's choice and the results of the round are displayed.

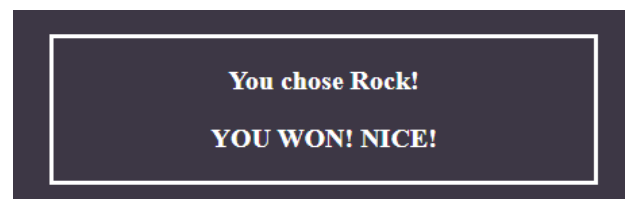


Figure 3: Results Displayed

5. Conclusion

This project was developed as an individual effort and helped reinforce the core programming concepts such as event handling, conditional logic, and dynamic UI updates. While the application is simple, it successfully demonstrates the fundamentals of web development.

In the future, additional features could be added, such as animations, sound effects, multiplayer functionality, and a more impressive UI. Additional potential improvements would be implementing a persistent scoring system using local database storage or possibly a cloud-based service with user login to track results after disconnecting.

References

[1]

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[2]

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